

Module 07: Communication — Neural Bases of Social Inference

Course: Active Inference 101 | **Unit:** Computational Neuroscience | **Audience:** First-semester undergraduates

Learning Objectives

1. Identify the **social brain network** — brain regions specialized for understanding other minds.
2. Explain how the **mirror neuron system** and **mentalizing network** support social inference.
3. Describe the neural basis of **language processing** as hierarchical predictive coding.

Introduction

Humans are intensely social. We infer intentions from glances, emotions from tone of voice, and beliefs from behavior. This module explores the brain circuits that make social communication possible — from recognizing faces to understanding sarcasm.

Key Concepts

1. The Social Brain Network

Several brain regions form a network dedicated to social cognition:

- **Superior temporal sulcus (STS):** Processes biological motion, gaze direction, and vocal prosody
- **Temporoparietal junction (TPJ):** Crucial for Theory of Mind — representing others' beliefs
- **Medial prefrontal cortex (mPFC):** Self-referential processing and reasoning about others' mental states
- **Fusiform face area (FFA):** Rapid face recognition

- **Amygdala:** Evaluating social salience and emotional significance

These regions work together to build a generative model of other agents.

2. Mirror Neuron System vs. Mentalizing Network

Two complementary systems support social understanding:

Mirror neuron system (premotor cortex + inferior parietal lobule):

- "Simulation route" — understanding others by internally simulating their actions
- Automatic and fast
- Good for understanding *what* someone is doing

Mentalizing network (TPJ + mPFC + temporal poles):

- "Theory route" — explicitly reasoning about others' beliefs and intentions
- Deliberate and slower
- Good for understanding *why* someone is doing something

Active Inference integrates both: the generative model uses simulation (mirror system) for rapid prediction and explicit reasoning (mentalizing) for deeper inference.

3. Language Processing as Predictive Coding

The brain processes language through a hierarchy similar to visual processing:

- **Auditory cortex:** Processes speech sounds (phonemes)
- **Wernicke's area** (left temporal): Processes word meanings (semantics)
- **Broca's area** (left frontal): Processes grammar (syntax) and generates speech predictions
- **Higher frontal regions:** Process discourse, context, and pragmatics

At each level, the brain predicts the next element:

- The **N400** ERP (negative deflection at 400ms) occurs when a word is semantically unexpected ("I take my coffee with cream and *socks*")
- The **P600** ERP occurs when syntax is violated ("The cat *were* sitting on the mat")

These ERPs are prediction error signals at different levels of the language hierarchy.

4. Oxytocin and Social Precision

Oxytocin is a neuropeptide that modulates social precision:

- Increases attention to social cues (faces, eyes, voices)
- Enhances trust and social bonding
- Increases the precision of social prediction errors → making social information "louder"

In Active Inference terms, oxytocin boosts the gain on social signals, making the agent more responsive to social information.

5. Autism Spectrum and Social Generative Models

Autism spectrum conditions may involve differences in social precision weighting:

- Difficulty predicting social behavior (high social prediction errors)
- Possible over-weighting of sensory precision and under-weighting of social priors
- This leads to the characteristic profile: strong detail-orientation but difficulty with social ambiguity

Summary

The social brain network (STS, TPJ, mPFC, FFA, amygdala) builds generative models of other agents. Mirror neurons provide fast action simulation, while the mentalizing network enables explicit Theory of Mind. Language is processed through predictive coding hierarchies, with N400 and P600 ERPs marking prediction errors. Oxytocin modulates social precision, and autism may involve differences in how social predictions are weighted.

Further Reading

- Frith, C. D. & Frith, U. (2012). Mechanisms of social cognition. *Annual Review of Psychology*, 63, 287-313.
- Hagoort, P. & Indefrey, P. (2014). The neurobiology of language beyond single words. *Annual Review of Neuroscience*, 37, 347-362.

- Lawson, R. P., Rees, G., & Friston, K. J. (2014). An aberrant precision account of autism. *Frontiers in Human Neuroscience*, 8, 302.